Social Media Engagement Analytics using Python

Project Overview

This project analyzes a dataset of **5,000 social media posts** to understand user engagement patterns, content performance, and audience behavior.

Using Python-based data analysis and visualization techniques, the project extracts insights that help improve content strategy, user targeting, and engagement optimization.

Objective

The main objectives of this project are:

- Analyze user engagement across post types and categories
- Identify factors influencing likes, comments, and shares
- Understand audience behavior and demographics
- Study engagement trends over time
- Perform statistical and correlation analysis

Dataset Description

- **Source:** Public dataset (GitHub)
- **Records:** 5000 posts
- **Features:** 20 columns

Key Attributes:

- User details: user_id, age, gender, country
- Post details: post_type, post_category, hashtags
- Engagement metrics: likes, comments, shares
- Performance metrics: impression_count, engagement_rate
- Behavioral data: watch_time, sentiment, follower_count

Tools & Technologies

- **Python**
- **Pandas** – Data cleaning & manipulation

- **NumPy** – Numerical analysis
- **Matplotlib & Seaborn** – Data visualization
- **Plotly** – Interactive visualizations

Data Cleaning & Preprocessing

- Handled missing values using median (numerical) and mode (categorical)
- Converted date columns to datetime format
- Removed inconsistencies and standardized categorical values
- Created new feature: hashtag_count
- Ensured correct data types for analysis

Feature Engineering

- Created **engagement_score = likes + comments + shares**
- Applied log transformation for skewed data
- Extracted date features for time-series analysis
- Derived engagement metrics for better insights

Analysis Performed

◆ **Engagement Analysis**

- Compared engagement across post types and categories
- Identified high-performing content types

◆ **Sentiment Analysis**

- Analyzed engagement based on sentiment
- Positive sentiment posts showed higher engagement

◆ **Geographic Analysis**

- Compared engagement across countries
- Some regions showed significantly higher interaction rates

◆ **Device Analysis**

- Studied engagement behavior based on device type

- Device type impacts interaction levels

🔹 **Time-Series Analysis**

- Identified engagement trends over time
- Detected fluctuations in user activity

🔹 **Correlation Analysis**

- Studied relationship between likes, comments, shares, and impressions
- Strong correlation between impressions and engagement

Statistical Analysis

- Calculated mean, median, and mode for engagement metrics
- Measured variance and standard deviation
- Identified skewness and data distribution patterns

Key Insights

- 📌 Higher impressions lead to higher engagement
- 📌 Post type significantly impacts performance
- 📌 Positive sentiment content performs better
- 📌 Certain countries show stronger engagement levels
- 📌 Hashtag usage improves content visibility
- 📌 Device type influences user interaction
- 📌 Engagement fluctuates over time

Business Insights

- Improve content strategy based on high-performing post types
- Focus on positive sentiment content for better engagement
- Use hashtags strategically for visibility
- Target high-engagement countries for marketing campaigns
- Optimize content for mobile and device behavior

Visualizations Used

Static Visualizations:

- Scatter Plot (Likes vs Impressions)
- Line Chart (Engagement Trends)
- Bar Chart (Post Categories)
- Histogram (Age Distribution)
- Box Plot (Engagement Rate)
- Heatmap (Correlation Matrix)

Interactive Visualizations:

- Plotly Line Charts
- Interactive Scatter Plots
- Bubble Charts
- Dynamic Bar Charts

Conclusion

This project demonstrates how Python can be used to analyze large-scale social media data and extract meaningful insights. The analysis helps understand user engagement behavior, optimize content strategy, and improve marketing decisions through data-driven insights.

Skills Demonstrated

- Data Cleaning & Preprocessing
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Data Visualization (Static & Interactive)
- Business Insight Generation
- Python Data Analysis Workflow